Prime Performance Evaluation

Report 5

Source Code and GitHub Repositories

The files used for this evaluation can be found at the following repository:
https://github.com/sam444555/cs2910

The original Prime code used is available at the following Johns Hopkins DSN repository:
https://github.com/jhu-dsn/prime

Aster Tests

- Single Driver Configuration
- No bandwidth limitation
- Unencrypted
- 25k total updates per trial
- 3 trials
- Hosts: aster13 → aster18
- Number of Emulated Clients: [20,30,50,75,100,150,200,300,500,750]
- MAX_PO_IN_FLIGHT Values: [20,50,100,250,500,750,1000]
- 210 runs, 5.25 million updates

MAX_PO_IN_FLIGHT: 20 Test Averages

Emulated	Time	Stalled Sends	Expected TPS	Actual TPS	Min Latency	Max Latency	Avg Latency
20	25.097	0.000	999.593	996.053	17.321	39.858	20.008
30	25.090	0.000	998.980	996.437	17.675	53.217	30.031
50	25.097	0.000	998.117	996.077	24.018	79.542	50.094
75	25.100	0.000	998.220	996.090	23.731	93.780	75.134
100	25.100	0.000	998.427	996.013	25.157	120.375	100.157
150	25.090	0.000	999.733	996.380	23.389	174.021	150.041
200	25.103	0.000	1,000.000	995.787	25.023	220.745	200.000
300	25.083	0.000	1,002.900	996.620	20.135	314.838	299.133
500	25.093	3,356,796.000	1,006.847	996.300	27.290	521.492	496.602
750	25.093	9,450,697.667	1,011.987	996.320	26.944	781.472	741.116

MAX_PO_IN_FLIGHT: 50 Test Averages

Emulated	Time	Stalled Sends	Expected TPS	Actual TPS	Min Latency	Max Latency	Avg Latency
20	25.097	0.000	999.803	996.157	16.318	39.721	20.004
30	16.743	0.000	1,499.483	1,493.223	17.278	33.658	20.007
50	10.293	0.000	2,439.973	2,428.763	17.536	51.300	20.492
75	10.087	0.000	2,487.517	2,478.450	17.989	67.198	30.150
100	10.067	0.000	2,492.287	2,483.083	20.574	73.580	40.124
150	10.053	0.000	2,495.680	2,486.577	33.368	78.533	60.104
200	10.077	0.000	2,494.193	2,480.580	24.543	115.400	80.187
300	10.050	0.000	2,504.800	2,488.260	27.194	134.940	119.770
500	10.077	2,257,131.000	2,510.097	2,481.127	28.688	235.956	199.198
750	10.050	8,350,870.667	2,530.317	2,488.140	30.278	316.154	296.406

MAX_PO_IN_FLIGHT: 100 Test Averages

Emulated	Time	Stalled Sends	Expected TPS	Actual TPS	Min Latency	Max Latency	Avg Latency
20	25.090	0.000	999.910	996.447	17.501	35.956	20.002
30	16.760	0.000	1,497.517	1,491.497	17.919	52.492	20.033
50	10.267	0.000	2,446.607	2,434.647	16.757	49.206	20.437
75	6.990	0.000	3,593.230	3,578.190	16.178	44.550	20.876
100	5.467	0.000	4,607.170	4,577.213	15.876	53.351	21.713
150	5.040	0.000	4,991.193	4,958.627	17.024	43.810	30.053
200	5.063	0.000	4,973.810	4,938.127	28.078	60.809	40.212
300	5.023	0.000	5,015.363	4,972.940	26.155	65.642	59.816
500	5.093	572,723.667	4,974.417	4,908.533	31.969	156.254	100.548
750	5.040	6,541,775.667	5,060.277	4,960.177	33.574	164.610	148.214

MAX_PO_IN_FLIGHT: 250 Test Averages

Emulated	Time	Stalled Sends	Expected TPS	Actual TPS	Min Latency	Max Latency	Avg Latency
20	25.117	0.000	999.007	995.273	17.035	39.934	20.020

30	16.767	0.000	1,497.480	1,491.437	17.771	37.224	20.034
50	10.280	0.000	2,441.753	2,432.537	17.298	51.617	20.484
75	6.993	0.000	3,595.060	3,575.407	16.319	43.001	20.863
100	5.707	0.000	4,406.463	4,385.473	16.268	55.571	22.714
150	4.230	0.000	5,956.283	5,923.240	14.692	40.046	25.213
200	3.823	0.000	6,581.067	6,540.027	16.015	57.380	30.396
300	2.767	0.000	9,145.043	9,039.207	13.790	52.865	32.810
500	2.273	287.000	11,278.620	11,012.193	30.793	107.078	44.399
750	2.743	1,663,922.333	9,324.227	9,113.600	40.802	117.266	80.461

MAX_PO_IN_FLIGHT: 500 Test Averages

Emulated	Time	Stalled Sends	Expected TPS	Actual TPS	Min Latency	Max Latency	Avg Latency
20	25.110	0.000	999.247	995.660	17.074	35.334	20.015
30	16.753	0.000	1,498.337	1,492.307	14.896	39.814	20.022
50	10.167	0.000	2,469.677	2,459.257	17.460	51.239	20.246
75	7.023	0.000	3,577.670	3,558.980	17.210	45.665	20.964
100	5.387	0.000	4,664.003	4,643.170	15.125	50.784	21.446
150	3.980	0.000	6,318.890	6,281.930	12.729	39.663	23.740
200	3.883	0.000	6,483.433	6,438.200	14.256	60.723	30.860
300	2.827	0.000	8,946.653	8,840.927	14.779	43.077	33.533
500	1.990	0.333	12,800.903	12,577.263	15.582	101.980	39.110
750	2.123	460,809.667	12,065.527	11,775.753	24.062	85.454	62.193

MAX_PO_IN_FLIGHT: 750 Test Averages

Emulated	Time	Stalled Sends	Expected TPS	Actual TPS	Min Latency	Max Latency	Avg Latency
20	25.103	0.000	999.263	995.790	15.369	37.121	20.015
30	16.730	0.000	1,500.530	1,494.297	16.717	29.788	19.993
50	10.203	0.000	2,461.113	2,449.880	17.301	46.239	20.317
75	7.110	0.000	3,533.677	3,519.700	17.144	44.159	21.245
100	5.967	0.000	4,211.710	4,189.507	16.025	54.876	23.744

150	4.067	0.000	6,183.243	6,152.080	14.504	38.691	24.273
200	3.443	0.000	7,310.920	7,263.463	11.436	57.453	27.365
300	2.793	0.000	9,066.280	8,953.800	15.487	42.321	33.096
500	2.003	737.000	12,703.580	12,498.593	19.900	101.812	39.410
750	2.127	589,397.333	12,035.827	11,765.770	21.155	82.082	62.316

MAX_PO_IN_FLIGHT: 1000 Test Averages

Emulated	Time	Stalled Sends	Expected TPS	Actual TPS	Min Latency	Max Latency	Avg Latency
20	25.103	0.000	999.477	995.847	14.553	39.663	20.011
30	16.757	0.000	1,498.337	1,492.197	17.522	39.165	20.022
50	10.120	0.000	2,480.013	2,470.067	16.812	50.727	20.162
75	6.903	0.000	3,637.943	3,620.610	15.739	40.819	20.622
100	5.310	0.000	4,728.667	4,707.270	14.863	53.147	21.148
150	4.057	0.000	6,202.517	6,164.337	13.017	36.849	24.190
200	3.660	0.000	6,891.883	6,834.200	14.713	58.949	29.043
300	2.807	0.000	8,990.043	8,902.000	16.010	42.836	33.373
500	1.963	210.000	13,014.953	12,783.350	18.146	99.862	38.518
750	2.163	705,652.333	11,847.607	11,574.977	21.921	84.962	63.337

(ASTER) Single Driver Configuration (Unlimited Mbps, 25k total sends, MAX_PO_IN_FLIGHT=20)							
Number of Emulated Clients: 20, Total Tests: 3							
Test #	Elapsed Time (seconds)	Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	25.09	0	1000.13	996.32	17.953	39.278	19.997
2	25.09	0	999.67	996.28	17.18	40.162	20.007
3	25.11	0	998.98	995.56	16.831	40.133	20.02
avg.	25.097	0	999.593	996.053	17.321	39.858	20.008
Number of Emulated Clients: 30, Total Tests: 3							
Test #	Elapsed Time (seconds)	Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	25.07	0	999.94	997.39	17.776	40.638	30.002
2	25.1	0	998.51	995.97	19.24	59.525	30.045
3	25.1	0	998.49	995.95	16.009	59.487	30.045
avg.	25.09	0	998.98	996.437	17.675	53.217	30.031
Number of Emulated Clients: 50, Total Tests: 3							
Test #	Elapsed Time (seconds)	Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	25.09	0	998.33	996.3	25.777	79.499	50.084
2	25.12	0	997.28	995.24	31.542	79.524	50.136
3	25.08	0	998.74	996.69	14.735	79.602	50.063
avg.	25.097	0	998.117	996.077	24.018	79.542	50.094
Number of Emulated Clients: 75, Total Tests: 3							

Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	25.12		0	997.36	995.23	32.382	100.018	75.199
2	25.07		0	999.49	997.35	18.807	81.35	75.039
3	25.11		0	997.81	995.69	20.003	99.971	75.165
avg.	25.1		0	998.22	996.09	23.731	93.78	75.134
Number of Emulated Clients: 100, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	25.1		0	998.52	996.11	29.698	120.301	100.148
2	25.11		0	997.93	995.52	23.603	120.422	100.207
3	25.09		0	998.83	996.41	22.169	120.401	100.117
avg.	25.1		0	998.427	996.013	25.157	120.375	100.157
Number of Emulated Clients: 150, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	25.11		0	999.19	995.69	20.152	179.831	150.122
2	25.07		0	1000.31	997.03	28.029	162.428	149.954
3	25.09		0	999.7	996.42	21.985	179.805	150.046
avg.	25.09		0	999.733	996.38	23.389	174.021	150.041
Number of Emulated Clients: 200, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	25.11		0	999.67	995.46	26.921	220.915	200.066
2	25.09		0	1000.54	996.32	24.845	220.672	199.893

3	25.11	0	999.79	995.58	23.304	220.647	200.042
avg.	25.103	0	1000	995.787	25.023	220.745	200
Number of Emulated Clients: 300, Total Tests: 3							
Test #	Elapsed Time (seconds)	Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	25.08	0	1002.93	996.63	19.941	320.878	299.124
2	25.1	0	1002.14	995.87	20.56	320.929	299.36
3	25.07	0	1003.63	997.36	19.903	302.708	298.916
avg.	25.083	0	1002.9	996.62	20.135	314.838	299.133
Number of Emulated Clients: 500, Total Tests: 3							
Test #	Elapsed Time (seconds)	Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	25.09	3356776	1006.8	996.26	24.773	521.461	496.625
2	25.09	3356816	1007.14	996.57	25.665	521.487	496.457
3	25.1	3356796	1006.6	996.07	31.433	521.529	496.723
avg.	25.093	3356796	1006.847	996.3	27.29	521.492	496.602
Number of Emulated Clients: 750, Total Tests: 3							
Test #	Elapsed Time (seconds)	Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	25.09	9450631	1012.28	996.6	24.829	781.455	740.901
2	25.1	9450711	1011.7	996.06	32.05	781.547	741.324
3	25.09	9450751	1011.98	996.3	23.954	781.415	741.122
avg.	25.093	9450697.667	1011.987	996.32	26.944	781.472	741.116
(ASTER) Single Driver Configuration (Unlimited Mbps, 25k total sends, MAX_PO_IN_FLIGHT=50)							
Number of Emulated Clients: 20, Total Tests: 3							

Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	25.09		0	1000.68	996.57	16.921	38.882	19.986
2	25.1		0	999.44	996.04	16.449	40.133	20.011
3	25.1		0	999.29	995.86	15.585	40.149	20.014
avg.	25.097		0	999.803	996.157	16.318	39.721	20.004
Number of Emulated Clients: 30, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	16.73		0	1500.76	1494.76	18.336	38.424	19.99
2	16.77		0	1496.92	1490.45	15.939	40.059	20.041
3	16.73		0	1500.77	1494.46	17.56	22.492	19.99
avg.	16.743		0	1499.483	1493.223	17.278	33.658	20.007
Number of Emulated Clients: 50, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	10.29		0	2441.47	2430.16	17.985	56.433	20.479
2	10.27		0	2446.45	2433.71	16.986	57.328	20.438
3	10.32		0	2432	2422.42	17.636	40.139	20.559
avg.	10.293		0	2439.973	2428.763	17.536	51.3	20.492
Number of Emulated Clients: 75, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency

1	10.1		0	2484.51	2476.28	18.83	64.599	30.187
2	10.06		0	2492.73	2484.9	18.463	58.486	30.087
3	10.1		0	2485.31	2474.17	16.675	78.509	30.177
avg.	10.087		0	2487.517	2478.45	17.989	67.198	30.15
Number of Emulated Clients: 100, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	10.03		0	2499.73	2492.11	23.94	42.02	40.004
2	10.11		0	2482.31	2472.33	24.031	118.534	40.285
3	10.06		0	2494.82	2484.81	13.75	60.187	40.083
avg.	10.067		0	2492.287	2483.083	20.574	73.58	40.124
Number of Emulated Clients: 150, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	10.04		0	2498.85	2489.81	33.604	67.157	60.028
2	10.04		0	2498.76	2489.59	34.616	68.188	60.03
3	10.08		0	2489.43	2480.33	31.883	100.255	60.255
avg.	10.053		0	2495.68	2486.577	33.368	78.533	60.104
Number of Emulated Clients: 200, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	10.1		0	2489.57	2474.33	16.655	157.741	80.335
2	10.05		0	2502.51	2488.29	20.629	88.163	79.92

3	10.08		0	2490.5	2479.12	36.345	100.297	80.305
avg.	10.077		0	2494.193	2480.58	24.543	115.4	80.187
Number of Emulated Clients: 300, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	10.04		0	2507.72	2491.13	28.223	122.207	119.63
2	10.03		0	2509.82	2493.22	21.54	122.168	119.53
3	10.08		0	2496.86	2480.43	31.819	160.445	120.151
avg.	10.05		0	2504.8	2488.26	27.194	134.94	119.77
Number of Emulated Clients: 500, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	10.12		2256671	2499.28	2470.49	31.802	280.142	200.058
2	10.04		2257716	2519.41	2491.28	27.523	203.838	198.459
3	10.07		2257006	2511.6	2481.61	26.74	223.888	199.076
avg.	10.077		2257131	2510.097	2481.127	28.688	235.956	199.198
Number of Emulated Clients: 750, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	10.04		8351021	2531.23	2490.4	31.524	304.437	296.299
2	10.06		8351400	2530.11	2485.82	29.287	318.462	296.43
3	10.05		8350191	2529.61	2488.2	30.022	325.562	296.489
avg.	10.05		8350870.667	2530.317	2488.14	30.278	316.154	296.406

(ASTER) Single Driver Configuration (Unlimited Mbps, 25k total sends, MAX_PO_IN_FLIGHT=100)								
Number of Emulated Clients: 20, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	25.08		0	1000.28	996.88	18.601	28.076	19.994
2	25.09		0	999.96	996.37	15.75	39.686	20.001
3	25.1		0	999.49	996.09	18.151	40.106	20.01
avg.	25.09		0	999.91	996.447	17.501	35.956	20.002
Number of Emulated Clients: 30, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	16.76		0	1497.32	1491.24	17.579	58.689	20.036
2	16.77		0	1496.75	1490.47	17.951	40.068	20.043
3	16.75		0	1498.48	1492.78	18.227	58.719	20.02
avg.	16.76		0	1497.517	1491.497	17.919	52.492	20.033
Number of Emulated Clients: 50, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	10.22		0	2457.88	2445.19	17.082	47.705	20.343
2	10.26		0	2448.47	2437.14	18.056	59.841	20.421
3	10.32		0	2433.47	2421.61	15.132	40.073	20.547
avg.	10.267		0	2446.607	2434.647	16.757	49.206	20.437
Number of Emulated Clients: 75, Total Tests: 3								

Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	7.01		0	3579.96	3566.16	15.564	55.715	20.95
2	6.87		0	3657.19	3639.96	17.061	37.82	20.508
3	7.09		0	3542.54	3528.45	15.908	40.115	21.171
avg.	6.99		0	3593.23	3578.19	16.178	44.55	20.876
Number of Emulated Clients: 100, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	5.55		0	4536.41	4505.36	15.257	40.142	22.044
2	5.53		0	4554.56	4524.75	16.165	79.725	21.956
3	5.32		0	4730.54	4701.53	16.207	40.186	21.139
avg.	5.467		0	4607.17	4577.213	15.876	53.351	21.713
Number of Emulated Clients: 150, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	5.05		0	4985.37	4949.58	16.379	53.446	30.088
2	5.03		0	5003.07	4965.61	16.816	38.788	29.982
3	5.04		0	4985.14	4960.69	17.878	39.196	30.089
avg.	5.04		0	4991.193	4958.627	17.024	43.81	30.053
Number of Emulated Clients: 200, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency

1	5.06		0	4983.45	4938.89	27.663	59.069	40.133
2	5.1		0	4941.2	4904.24	23.169	75.355	40.476
3	5.03		0	4996.78	4971.25	33.402	48.002	40.026
avg.	5.063		0	4973.81	4938.127	28.078	60.809	40.212
Number of Emulated Clients: 300, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	5.01		0	5021.95	4985.61	20.073	61.525	59.738
2	5.04		0	5010.93	4956.7	29.274	74.142	59.869
3	5.02		0	5013.21	4976.51	29.118	61.259	59.842
avg.	5.023		0	5015.363	4972.94	26.155	65.642	59.816
Number of Emulated Clients: 500, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	5.23		573193	4846.62	4781.62	33.697	257.47	103.165
2	5.02		571636	5037.18	4975.27	30.447	104.82	99.262
3	5.03		573342	5039.45	4968.71	31.763	106.471	99.217
avg.	5.093		57272 3.667	4974.417	4908.533	31.969	156.254	100.548
Number of Emulated Clients: 750, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	5.03		6541081	5072.44	4966.3	34.406	158.821	147.858
2	5.03		6542419	5063.42	4974.75	30.846	157.874	148.121

3	5.06		6541827	5044.97	4939.48	35.47	177.136	148.663
avg.	5.04		6541775.667	5060.277	4960.177	33.574	164.61	148.214
(ASTER) Single Driver Configuration (Unlimited Mbps, 25k total sends, MAX_PO_IN_FLIGHT=250)								
Number of Emulated Clients: 20, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	25.13		0	998.85	994.7	17.592	40.061	20.023
2	25.11		0	998.85	995.46	17.624	40.149	20.023
3	25.11		0	999.32	995.66	15.89	39.592	20.014
avg.	25.117		0	999.007	995.273	17.035	39.934	20.02
Number of Emulated Clients: 30, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	16.79		0	1494.84	1489.28	17.69	40.065	20.069
2	16.75		0	1499.25	1492.96	17.688	31.552	20.01
3	16.76		0	1498.35	1492.07	17.935	40.054	20.022
avg.	16.767		0	1497.48	1491.437	17.771	37.224	20.034
Number of Emulated Clients: 50, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	10.49		0	2391.76	2382.58	17.504	56.008	20.905
2	10.3		0	2436.19	2426.59	17.697	58.784	20.524
3	10.05		0	2497.31	2488.44	16.694	40.06	20.022
avg.	10.28		0	2441.753	2432.537	17.298	51.617	20.484

Number of Emulated Clients: 75, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	7.04		0	3572.26	3551.19	15.916	48.645	20.995
2	6.92		0	3629.16	3612.93	17.271	40.126	20.666
3	7.02		0	3583.76	3562.1	15.769	40.233	20.928
avg.	6.993		0	3595.06	3575.407	16.319	43.001	20.863
Number of Emulated Clients: 100, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	5.63		0	4462.37	4443.12	16.4	37.472	22.41
2	5.54		0	4529.56	4509.37	15.209	90.648	22.077
3	5.95		0	4227.46	4203.93	17.196	38.593	23.655
avg.	5.707		0	4406.463	4385.473	16.268	55.571	22.714
Number of Emulated Clients: 150, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	4.25		0	5925.58	5888.89	14.554	40.035	25.314
2	4.4		0	5724.34	5688.14	16.64	40.128	26.204
3	4.04		0	6218.93	6192.69	12.881	39.974	24.12
avg.	4.23		0	5956.283	5923.24	14.692	40.046	25.213
Number of Emulated Clients: 200, Total Tests: 3								

Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	3.86		0	6530.31	6471.09	15.884	40.646	30.626
2	3.75		0	6710.28	6673.82	16.952	91.458	29.805
3	3.86		0	6502.61	6475.17	15.208	40.036	30.757
avg.	3.823		0	6581.067	6540.027	16.015	57.38	30.396
Number of Emulated Clients: 300, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	2.74		0	9203.92	9117.73	15.286	48.706	32.595
2	2.74		0	9252.23	9130.58	12.944	55.453	32.425
3	2.82		0	8978.98	8869.31	13.14	54.435	33.411
avg.	2.767		0	9145.043	9039.207	13.79	52.865	32.81
Number of Emulated Clients: 500, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	2.35		861	10872.97	10637.69	29.812	205.498	45.986
2	2.16		0	11896.42	11588.6	32.976	60.044	42.029
3	2.31		0	11066.47	10810.29	29.59	55.691	45.182
avg.	2.273		287	11278.62	11012.193	30.793	107.078	44.399
Number of Emulated Clients: 750, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency

1	2.68		15150 65	9547.6 6	9321.53	29.17 8		
(ASTER) Single Driver Configuration (Unlimited Mbps, 25k total sends, MAX_PO_IN_FLIGHT=500)								
Number of Emulated Clients: 20, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expect ed TPS	Actual Throughp ut	Min Laten cy	Max Latenc y	Avg Laten cy
1	25.12		0	999.33	995.35	18.98 2	40.04 6	20.01 3
2	25.09		0	999.68	996.29	15.79 7	30.17 5	20.00 6
3	25.12		0	998.73	995.34	16.44 4	35.78	20.02 5
avg.	25.11		0	999.24 7	995.66	17.07 4	35.33 4	20.01 5
Number of Emulated Clients: 30, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expect ed TPS	Actual Throughp ut	Min Laten cy	Max Latenc y	Avg Laten cy
1	16.74		0	1499.5 7	1493.56	11.27 7	38.68	20.00 6
2	16.78		0	1496.4	1489.73	15.10 5	41.70 2	20.04 8
3	16.74		0	1499.0 4	1493.63	18.30 5	39.06 1	20.01 3
avg.	16.753		0	1498.3 37	1492.307	14.89 6	39.81 4	20.02 2
Number of Emulated Clients: 50, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expect ed TPS	Actual Throughp ut	Min Laten cy	Max Latenc y	Avg Laten cy
1	10.19		0	2463.8 6	2453.94	17.48 5	56.27 7	20.29 3
2	10.19		0	2463.1 3	2453.3	17.70 4	57.43 5	20.29 9
3	10.12		0	2482.0 4	2470.53	17.19	40.00 4	20.14 5
avg.	10.167		0	2469.6 77	2459.257	17.46	51.23 9	20.24 6
Number of Emulated Clients: 75, Total Tests: 3								

Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	7.09		0	3544.31	3525.94	17.735	60.669	21.161
2	7.01		0	3583.82	3566.13	17.404	38.935	20.927
3	6.97		0	3604.88	3584.87	16.49	37.392	20.805
avg.	7.023		0	3577.67	3558.98	17.21	45.665	20.964
Number of Emulated Clients: 100, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	5.47		0	4590.32	4572.88	14.575	37.112	21.785
2	5.41		0	4642.02	4617.73	16.479	79.434	21.542
3	5.28		0	4759.67	4738.9	14.321	35.807	21.01
avg.	5.387		0	4664.003	4643.17	15.125	50.784	21.446
Number of Emulated Clients: 150, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	3.96		0	6339.04	6306.7	12.75	39.859	23.663
2	3.95		0	6360.06	6330.26	10.869	39.137	23.585
3	4.03		0	6257.57	6208.83	14.567	39.993	23.971
avg.	3.98		0	6318.89	6281.93	12.729	39.663	23.74
Number of Emulated Clients: 200, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency

1	3.84		0	6551.11	6502.71	16.823	40.689	30.529
2	4		0	6305.19	6249.5	13.588	100.814	31.72
3	3.81		0	6594	6562.39	12.356	40.666	30.331
avg.	3.883		0	6483.433	6438.2	14.256	60.723	30.86
Number of Emulated Clients: 300, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	2.82		0	8978.85	8861.58	11.767	43.217	33.412
2	2.84		0	8898.62	8794.16	16.478	43.797	33.713
3	2.82		0	8962.49	8867.04	16.092	42.216	33.473
avg.	2.827		0	8946.653	8840.927	14.779	43.077	33.533
Number of Emulated Clients: 500, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	2.08		0	12175.74	12006.86	15.433	200.811	41.065
2	1.97		1	12975.43	12703.9	14.496	54.34	38.534
3	1.92		0	13251.54	13021.03	16.818	50.788	37.731
avg.	1.99		0.333	12800.903	12577.263	15.582	101.98	39.11
Number of Emulated Clients: 750, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	2.06		377455	12437.59	12152.66	23.846	79.922	60.301

2	2.18		498304	11768.62	11448.5	23.395	84.719	63.729
3	2.13		506670	11990.37	11726.1	24.944	91.721	62.55
avg.	2.123		460809.667	12065.527	11775.753	24.062	85.454	62.193
(ASTER) Single Driver Configuration (Unlimited Mbps, 25k total sends, MAX_PO_IN_FLIGHT=750)								
Number of Emulated Clients: 20, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	25.08		0	1000.15	996.77	17.204	31.139	19.997
2	25.11		0	998.87	995.48	18.065	40.059	20.023
3	25.12		0	998.77	995.12	10.838	40.164	20.025
avg.	25.103		0	999.263	995.79	15.369	37.121	20.015
Number of Emulated Clients: 30, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	16.72		0	1501.48	1495.16	15.061	22.231	19.98
2	16.75		0	1498.71	1492.65	17.945	44.56	20.017
3	16.72		0	1501.4	1495.08	17.145	22.572	19.981
avg.	16.73		0	1500.53	1494.297	16.717	29.788	19.993
Number of Emulated Clients: 50, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	10.1		0	2485.11	2474.22	16.88	55.063	20.12
2	10.3		0	2437.65	2427.48	17.897	44.597	20.512
3	10.21		0	2460.58	2447.94	17.125	39.058	20.32

avg.	10.203		0	2461.113	2449.88	17.301	46.239	20.317
Number of Emulated Clients: 75, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	7		0	3585.07	3571.56	19.094	50.013	20.92
2	7.42		0	3381.45	3367.37	16.66	41.24	22.18
3	6.91		0	3634.51	3620.17	15.678	41.224	20.635
avg.	7.11		0	3533.677	3519.7	17.144	44.159	21.245
Number of Emulated Clients: 100, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	5.93		0	4241.73	4216.36	15.798	37.51	23.575
2	6.02		0	4173.02	4150.4	16.592	86.881	23.963
3	5.95		0	4220.38	4201.76	15.684	40.238	23.695
avg.	5.967		0	4211.71	4189.507	16.025	54.876	23.744
Number of Emulated Clients: 150, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	4.2		0	5985.96	5954.67	15.322	39.9	25.059
2	4.03		0	6235.99	6202.62	13.596	39.861	24.054
3	3.97		0	6327.78	6298.95	14.593	36.313	23.705
avg.	4.067		0	6183.243	6152.08	14.504	38.691	24.273

Number of Emulated Clients: 200, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	3.37		0	7490.39	7426.61	11.983	39.883	26.701
2	3.49		0	7200.67	7157.8	12.318	93.167	27.775
3	3.47		0	7241.7	7205.98	10.007	39.31	27.618
avg.	3.443		0	7310.92	7263.463	11.436	57.453	27.365
Number of Emulated Clients: 300, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	2.74		0	9224.96	9137.74	16.945	44.01	32.52
2	2.84		0	8923.71	8795.04	15.661	40.037	33.618
3	2.8		0	9050.17	8928.62	13.856	42.915	33.149
avg.	2.793		0	9066.28	8953.8	15.487	42.321	33.096
Number of Emulated Clients: 500, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	2.1		2211	12081.09	11904.72	19.368	206.854	41.387
2	1.94		0	13149.36	12914.21	23.219	48.804	38.025
3	1.97		0	12880.29	12676.85	17.113	49.777	38.819
avg.	2.003		737	12703.58	12498.593	19.9	101.812	39.41
Number of Emulated Clients: 750, Total Tests: 3								

Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	2.12		617120	12092.58	11806.02	27.368	80.617	62.022
2	2.15		646595	11933.15	11637.27	11.263	83.833	62.85
3	2.11		504477	12081.75	11854.02	24.834	81.797	62.077
avg.	2.127		589397.333	12035.827	11765.77	21.155	82.082	62.316
(ASTER) Single Driver Configuration (Unlimited Mbps, 25k total sends, MAX_PO_IN_FLIGHT=1k)								
Number of Emulated Clients: 20, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	25.08		0	1000.95	996.83	10.977	38.799	19.981
2	25.09		0	999.76	996.37	15.582	40.08	20.005
3	25.14		0	997.72	994.34	17.099	40.109	20.046
avg.	25.103		0	999.477	995.847	14.553	39.663	20.011
Number of Emulated Clients: 30, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	16.75		0	1499.33	1492.79	18.28	39.787	20.009
2	16.78		0	1495.74	1490.17	17.482	45.652	20.057
3	16.74		0	1499.94	1493.63	16.804	32.057	20.001
avg.	16.757		0	1498.337	1492.197	17.522	39.165	20.022
Number of Emulated Clients: 50, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	10.16		0	2468.21	2459.49	16.685	55.724	20.258

2	10.13		0	2478.18	2468.37	16.881	58.822	20.176
3	10.07		0	2493.65	2482.34	16.869	37.635	20.051
avg.	10.12		0	2480.013	2470.067	16.812	50.727	20.162
Number of Emulated Clients: 75, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	7.05		0	3566.88	3544.97	15.487	45.744	21.027
2	6.9		0	3635.69	3620.67	16.1	40.115	20.629
3	6.76		0	3711.26	3696.19	15.631	36.598	20.209
avg.	6.903		0	3637.943	3620.61	15.739	40.819	20.622
Number of Emulated Clients: 100, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	5.29		0	4743.57	4724.89	13.433	40.06	21.081
2	5.33		0	4709.34	4686.69	15.571	79.333	21.234
3	5.31		0	4733.09	4710.23	15.584	40.048	21.128
avg.	5.31		0	4728.667	4707.27	14.863	53.147	21.148
Number of Emulated Clients: 150, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	4.15		0	6061.95	6019.47	13.603	36.926	24.745
2	4.02		0	6267.53	6224.68	13.91	37.317	23.933
3	4		0	6278.07	6248.86	11.538	36.305	23.893

avg.	4.057		0	6202.517	6164.337	13.017	36.849	24.19
Number of Emulated Clients: 200, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	3.57		0	7065.72	6998.23	15.093	40.103	28.306
2	3.81		0	6620.44	6566.67	16.261	40.163	30.209
3	3.6		0	6989.49	6937.7	12.784	96.581	28.614
avg.	3.66		0	6891.883	6834.2	14.713	58.949	29.043
Number of Emulated Clients: 300, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	2.77		0	9108.42	9032.85	16.596	42.969	32.937
2	2.82		0	8931.71	8850.06	15.69	42.016	33.588
3	2.83		0	8930	8823.09	15.743	43.523	33.595
avg.	2.807		0	8990.043	8902	16.01	42.836	33.373
Number of Emulated Clients: 500, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	2.1		630	12091.21	11916.39	17.043	206.559	41.352
2	1.89		0	13482.67	13254.82	19.591	46.4	37.085
3	1.9		0	13470.98	13178.84	17.805	46.627	37.117
avg.	1.963		210	13014.953	12783.35	18.146	99.862	38.518

Number of Emulated Clients: 750, Total Tests: 3								
Test #	Elapsed Time (seconds)		Stalled Sends	Expected TPS	Actual Throughput	Min Latency	Max Latency	Avg Latency
1	2.09		585393	12190.92	11972.47	19.84	80.782	61.521
2	2.23		792064	11523.85	11219.21	25.64	88.015	65.082
3	2.17		739500	11828.05	11533.25	20.283	86.089	63.409
avg.	2.163		705652.333	11847.607	11574.977	21.921	84.962	63.337